

HISTORY TO FUTURE : Evolving Agent with Experience and Thought for Zero-shot Vision-and-Language Navigation

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“The only source of knowledge is experience. All knowledge of reality starts from experience and ends in it.”

— Albert Einstein

Abstract

Vision-and-Language Navigation in Continuous Environment (VLN-CE) requires an agent to follow language instructions to navigate the target destination. With the advancement of large language models (LLMs), recent efforts have explored adapting them for zero-shot VLN-CE, offering a promising solution in addressing the drawbacks of poor generalization in the training-based paradigm. However, existing LLM-based works primarily perform naive reasoning for decision-making and lack feedback, e.g., reviewing historical errors and predicting future potentials. Consequently, it may suffer from continuous failure for those initial error tasks. In this paper, we rethink LLM-based zero-shot VLN-CE and propose a new paradigm, named EVONAV, to improve the agent’s decision-making with future thought and history experience via Future Chain-of-Thought (F-CoT) and History Chain-of-Experience (H-CoE). F-CoT predicts future actions and landmarks as thoughts to assist navigation progress estimation and direction selection, while H-CoE summarizes historical trajectories and scenes as experience to improve navigation decision reliability. Both F-CoT and H-CoE cooperatively evolve the agent’s decision-making. Extensive experiments in both the simulator and real-world environments demonstrate the effectiveness of our EVONAV.

1. Introduction

State-of-the-art embodied models [14, 47, 54] typically follow the training paradigm on task-specific datasets [6, 24,

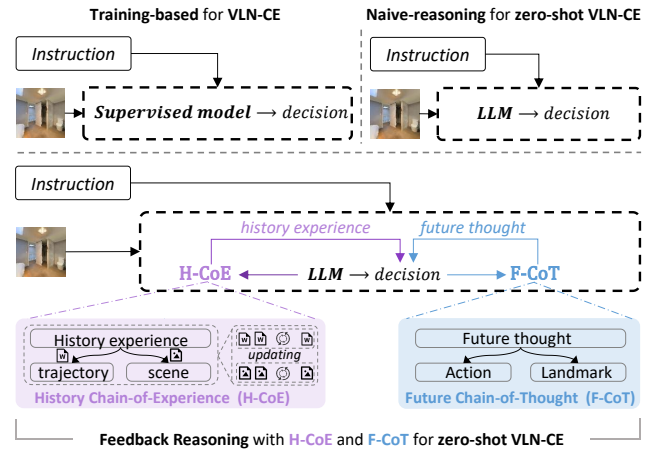


Figure 1. Previous methods typically focused on the task of VLN-CE under **Training-based** paradigms, suffering from poor zero-shot generalization in unseen scenes. Recent methods address this by LLMs in rich world knowledge and have explored adapting them for zero-shot VLN-CE, but incur **Naive reasoning** without feedback, e.g., reviewing historical errors and predicting future potentials, leading to poor decision-making. This paper improves LLM-based agents’ decision-making by the **Feedback Reasoning** of history experience and future thought via **History Chain-of-Experience (H-CoE)** and **Future Chain-of-Thought (F-CoT)**.

34] to achieve promising accuracy. However, such training-based paradigms require models to build a fixed inductive bias for certain seen environments, resulting in poor generalization in new scenarios. More importantly, these drawbacks of training-based paradigms (e.g., the top left of Figure 1) have become more prominent and urgent problems that need to be addressed for zero-shot embodied navigation tasks [35, 38, 39], particularly for zero-shot Vision-and-Language Navigation in Continuous Environment (VLN-CE) [35, 38, 39, 44, 53]. This task enables the agent to follow language instructions, flexibly explore in a continu-

ous environment, and navigate to the destination.

Recently, with the rapid development of large language models (LLMs) [1, 8, 18, 22], following powerful reasoning ability and rich commonsense knowledge, LLM-based paradigms have become a promising approach for embodied navigation. The emerging works have attempted to extend these versatile LLMs and adapt them for VLN-CE [2, 11, 26, 55, 57], achieving brilliant progress in addressing the drawbacks of poor generalization under the training-based paradigm. As shown in the top right of Figure 1, these works generally use multiple generalist visual translators (*e.g.*, BLIP-2 [28]) to translate visual scenes into universal linguistic text [13, 26] and then reason about navigation actions with rich world knowledge in LLMs.

Despite the significant progress achieved by LMMs, these methods use extra visual translators to indirectly process visual environments rather than using LLMs, leading to poor reasoning of visual details. More importantly, these methods simply perform naive reasoning [12, 35, 53] for decision-making and lack the feedback of backtracking history and predicting future, *e.g.*, reviewing errors of historical episodes and imagining potential results of future. Such feedback is more valuable, especially for those failure cases. Therefore, this leads to a question: *have we fully unleashed the potential of LLMs for zero-shot VLN-CE?*

In this paper, the answer is “NO”. The above paradigm relying on naive reasoning without feedback may suffer from heavy decision hallucination for those failed tasks. Consequently, it may encounter continuous failure for those initial navigation error tasks. In contrast, like humans in navigation, we can continuously evolve our decision-making from history to future feedback [7, 25], following *history* $\rightarrow$ *now* $\rightarrow$ *future*. Among them, in assisting current decision-making, we summarize historical errors to provide experience while imagining future results to avoid risk. In light of this, it inspires us to follow the human mechanism for evolving agents in navigation. Thus, *this paper rethink LLM-based zero-shot VLN-CE and propose a new framework, named EvoNav, to improve the agent’s decision-making by history experience and future thought.*

Specifically, as illustrated at the bottom of Figure 1, EvoNav addresses the above process by two insightful reasoning: ❶ **History Chain-of-Experience (H-CoE)** and ❷ **Future Chain-of-Thought (F-CoT)**. For ❶ H-CoE, it provides history experience from two complementary sources: *textual trajectory experience* and *visual scene experience*. Among them, the textual trajectory experience provides global guidance for navigation episodes, while the visual scene experience delivers local visual images for navigation step. For ❷ F-CoT, it provides future thought by predicting future actions and landmarks, which assist agent to track the current navigation progress and potential moving directions. The history experience and future thought are

cooperative to refine the agent’s decision-making. Both extensive experiments in the simulator and real-world environment demonstrate the effectiveness of our EvoNav.

In summary, our main contributions are:

- We rethink LLM-based zero-shot VLN-CE and present a new history-future reasoning paradigm, named **EVONAV**, which evolves the agent’s decision-making by backtracking history experience and predicting future thought.
- We present a **Future Chain-of-Thought (F-CoT)** reasoning, which predicts future actions and landmarks as thought to assist the agent in tracking current navigation progress and direction prediction.
- We present a **History Chain-of-Experience (H-CoE)** reasoning, which summarizes textual trajectories and visual scenes from historical episodes as experience to improve current decision-making.

For thorough examination (§5), we test EVONAV across several diverse open- and closed-source LMMs. Experimental results demonstrate that EVONAV outperforms the counterparts, *e.g.*, +20% SR, +21% OSR, and +17% SPL gains over R2R-CE [6], +6% SR, +7% OSR, and +4% SPL on more complex NavRAG-CE [48], which requires agents deeper understanding of user demands. EVONAV not only helps release the reasoning power of existing LLMs, but also evidences the merits of historical experience and future thought methods in challenging VLN-CE decision-making. Source code will be released.

2. Related Work

2.1. Vision-and-Language Navigation

Vision-and-Language Navigation (VLN) tasks involve agents navigating to target locations based on natural language instructions. Traditional approaches primarily used discrete settings with the MP3D simulator [9], where navigation was based on predefined graphs and fixed node selections. To enable more realistic interactions, the continuous environment setting (VLN-CE) was introduced, allowing agents to move freely using low-level actions. Initial VLN-CE methods employed encoder-decoder architectures [6] with LSTMs and attention [3, 16, 19, 27, 32, 36, 40–43, 45, 49–52]. With the rise of vision Transformers [15], adapting Transformer models for VLN-CE became prevalent [5, 10, 21, 31, 33, 46]. Recent works fine-tuning large models, such as Navid [54] of scaling to 550k samples for training LLM, offer promising solutions. However, training VLN models on domain-specific data limits their zero-shot generalization on unseen environments. This paper focuses on the more challenging and practical setting of zero-shot VLN-CE, aiming to improve LLM-based agents’ decision-making by historical experience and future thought.

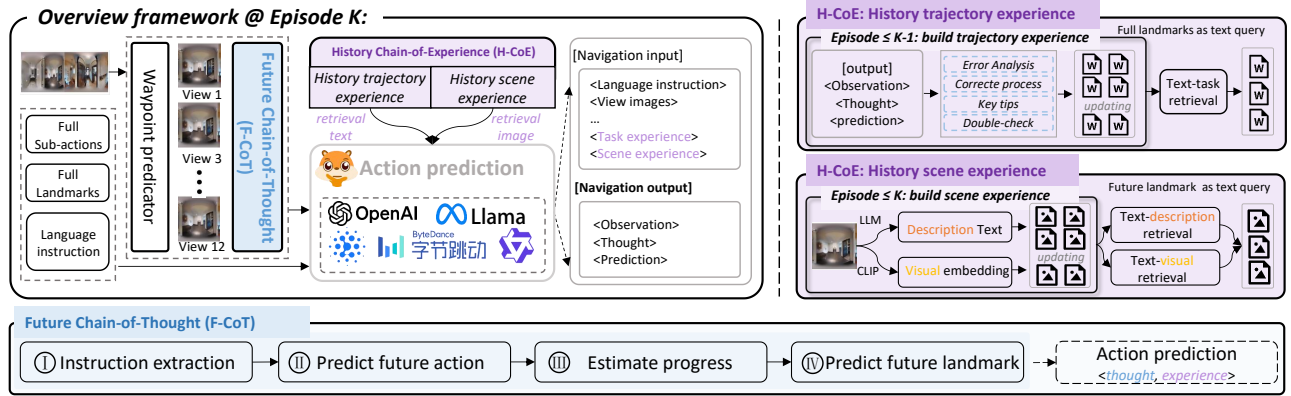


Figure 2. Overview framework of EvoNav. EvoNav addresses zero-shot VLN-CE with two complementary reasoning schemes: Future Chain-of-Thought (F-CoT) and History Chain-of-Experience (H-CoE). Given panoramic RGB and depth images, a standard waypoint predictor [20] first identifies candidate waypoint images. These images are then analyzed by the F-CoT, which predicts next-step actions and landmarks as thoughts to guide current navigation progress and direction. After that, H-CoE retrieves trajectory- and scene-related experience from a dynamic experience bank that continuously summarizes and stores completed trajectories and visited scenes. Finally, the retrieved historical experiences and future thoughts are fused within the action prediction module to support decision-making.

2.2. Large Language Models for Navigation

With the emergence of large language models (LLMs), recent works have explored their potential in VLN-CE. Some approaches use LLMs as navigation planners, such as DiscussNav [29], A²Nav [13], CA-Nav [12], and OpenNav [35]. To enhance LLM-based agents with navigation knowledge, some works scale domain-specific data or train additional navigation modules [38, 54, 58]. While these methods improve performance, they hinder sim-to-real transfer in VLN-CE, as they require extra data training. Additionally, these LLM-based works typically perform direct unidirectional reasoning for decision-making and lack feedback, they may fail continuously after an initial navigation error, resulting in poor decision-making. In contrast, this paper propose an EvoNav framework for zero-shot VLN-CE that improves the agent’s decision-making by history experience and future thought.

2.3. Lifelong Learning

The most recent works [17, 23, 47, 56] propose to maintain global memory to store environment 3D structure for embodied tasks, offering a promising solution in addressing embodied navigation with lifelong learning. Such a memory-based approach shares a similar spirit with us in evolving agents’ decision-making with history experience, but differs in key aspects. First, maintaining and updating such memories typically relies on complex localization tools (*e.g.*, SLAM [23, 47]) to construct environment representation, while our experience is free from these extra tools. Second, these methods primarily directly store geometric structure but lack task-related knowledge, while our experience improves its via task-related experience and thought. Third, such memories are useful only when the agent revisits the same environment, while EvoNav experi-

ence can generalize to new environments.

3. Problem Formulation

Vision-and-Language Navigation in Continuous Environments (VLN-CE) requires an agent to follow the language instruction and navigate to the target location with low-level actions, such as move forward 0.25m, turn left and right 15°. At each step, the agent receives panoramic RGB-D visuals $\mathcal{O}$, comprised of 12 RGB images and depth images surrounding its current location (*e.g.*, 0°, 30°, ..., 330°). In addition, the agent is provided with language instruction $\mathcal{I}$ for each episode. No pre-built maps are allowed in this task.

4. Method

4.1. Preliminary: LLM-based zero-shot VLN-CE

Existing LLM-based paradigms of zero-shot VLN-CE primarily simply associate the instruction $\mathcal{I}$ and current visual observations $\mathcal{O}$ with LLMs and perform direct reasoning to predict agent actions $\mathcal{A}_{now}$, as follows

$$\mathcal{LLM}(\mathcal{O}, \mathcal{I}) \rightarrow \mathcal{A}_{now}. \quad (1)$$

However, such a paradigm may suffer from heavy decision hallucination due to neglecting the reasoning feedback, *e.g.*, reviewing historical errors and predicting future potentials.

In this paper, we rethink zero-shot VLN-CE and present a novel paradigm, *aka* **EvoNav**, to continuously evolve the agent’s decision-making by leveraging reasoning of history experience and future thought. As a result, the Eq. 1 is reformulated as,

$$\begin{array}{c} \text{experience} \quad \quad \quad \text{thought} \\ \swarrow \quad \quad \quad \searrow \\ \text{H-CoE} \leftarrow \mathcal{LLM}(\mathcal{O}, \mathcal{I}) \rightarrow \mathcal{A}_{now} \rightarrow \text{F-CoT}. \end{array} \quad (2)$$

Overview Framework. As demonstrated in Figure 2, EvoNav mainly contains two complementary reasoning

scheme: i) *Future Chain-of-Thought* (F-CoT, §4.2) that predicts next-step actions and landmarks as thought to assist in tracking current navigation progress and direction; and ii) *History Chain-of-Experience* (H-CoE, §4.3) that summarizes historical task-level decisions and step-level scenes as experience to assist current navigation decisions.

4.2. Future Chain-of-Thought (F-CoT)

F-CoT is designed to assist in navigation progress estimation and direction selection with future thought by four main stages: instruction extraction, future action prediction, progress estimation, and future landmark prediction.

4.2.1. Instruction Extraction

To capture instruction details, we use LLM to parse the full instructions into two subsets, *i.e.*, temporal sub-actions and spatial landmarks. The sub-actions correspond to the movements (*e.g.*, “go through”) while spatial landmarks are objects (*e.g.*, “the white double doors”) in the instruction, as

$$(\mathcal{I}^t, \mathcal{I}^s) = \mathcal{LLM}(\mathcal{I}), \quad (3)$$

where $\mathcal{I}^s/\mathcal{I}^t$ denotes the parsed full sub-actions/landmarks.

4.2.2. Predict Future Action

To track navigation progress, we use an LLM to predict future actions $\mathcal{P}^f$. Note that since future actions have not yet occurred, the agent’s state can be inferred by excluding the predicted future action from completed sub-actions, as

$$\mathcal{P}^f = \mathcal{LLM}(\mathcal{H}, \mathcal{O}, \mathcal{I}), \quad (4)$$

where the $\mathcal{H}$ denotes the past step’s path information before the current step. Note that when the agent is in the start position, the $\mathcal{H}$ is set as “Step 0 start position”.

4.2.3. Estimate Progress

Based on the future action $\mathcal{P}^f$, we comprehensively review the historical finished path to infer current navigation progress, *i.e.*, have executed actions $\mathcal{A}^d$, as,

$$\mathcal{A}^d = \mathcal{LLM}(\mathcal{I}^s, \mathcal{I}^t, \mathcal{P}^f, \mathcal{H}). \quad (5)$$

4.2.4. Predict Future Landmark

To enhance the agent’s perception of the future direction, we prompt the LLM to infer future landmarks from all potential landmarks, by the following rules:

- *Rule 1-When the next action explicitly references a landmark:* i) Extract the mentioned landmark object; ii) Match it with an entry in the Landmarks List; iii) Output the landmark name exactly as recorded in the list.
- *Rule 2-When the next action only describes directional movement without a landmark reference:* Actions like “Turn left/right” “Go forward” contain no landmark information and should directly output “None.”

With above rules, the future landmark $\mathcal{L}^f$ is calculated as,

$$\mathcal{L}^f = \mathcal{LLM}(\mathcal{I}, \mathcal{I}^s, \mathcal{I}^t, \mathcal{A}^d). \quad (6)$$

4.3. History Chain-of-Experience (H-CoE)

Here, we elaborate a History Chain-of-Experience (H-CoE) reasoning, which online summarizes historical navigation task textual trajectories and visual images as experience, for improving the current decision-making process with similar instructions or scenes. This is achieved by two complementary historical experiences, the *textual trajectory experience* and the *visual scene experience*, as follows.

4.3.1. Textual Trajectory Experience

Metric Evaluation. For each finished episode, we online test the agent’s effective and efficient performance R with moving to the destination, formulated as,

$$R = \psi_{(NE, SR, OSR, SPL)}(Nav, Nav^{GT}), \quad (7)$$

where Nav and Nav^{GT} are predicted path and ground-truth path. The (NE, SR, OSR, SPL) denotes evaluation metrics, *please see more metric details in Sec. 5.1.*

Navigation Summary. Using R and the complete navigation trajectory $\mathcal{H}^*$ of each finished episode, we prompt LLM to summarize its decision process as the experience $\mathcal{E}$ to assist the decision-making for future similar episodes, as

- *Rule 1-Complete Failure (SR=0):* Carefully analyze the navigation error; Provide a corrected navigation process; Summarize key tips for similar tasks; and double-check.
- *Rule 2-Perfect Navigation (SR=1 and SPL ≥ 0.8):* Reproduce this decision process exactly as supported.
- *Rule 3-Successful but Inefficient (SR=1 and SPL < 0.8):* Provide optimized navigation suggestions for higher SPL.

Trajectory Experience Construction. With each finished episode’s trajectory experience $\mathcal{E}$, we use chroma (a vector database) to continuously store it in the experience bank $\mathcal{M}$. When the n -th episode is finished, the $\mathcal{M}_n$ is online constructed as,

$$\mathcal{M}_n = \mathcal{M}_{n-1} \cup [(\mathcal{I}_n, \mathcal{I}_n^s, \mathcal{H}_n^*, \mathcal{E}_n)]_{n=1}^N, \quad (8)$$

where N is the dataset volume. *Please Note that experience can only be collected after an episode is completed.* In other words, when the n -th episode is not finished, $\mathcal{M}$ stores only the experience from the previous $(n-1)$ episode.

Trajectory Experience Retrieval. Here, we use the full landmarks $\mathcal{I}^s$ of the current episode as the query to retrieve the top-K experiences from $\mathcal{M}$. For n -th episode, the retrieved trajectory experiences $\mathcal{T}_n^e$ is formulated as,

$$\mathcal{T}_n^e = T2T_{retrieval} \left(\frac{\mathcal{I}_n^s \cdot \mathcal{M}_{(n-1)}}{\|\mathcal{I}_n^s\| \cdot \|\mathcal{M}_{(n-1)}\|} \right), \quad (9)$$

where $T2T_{retrieval}$ is the retrieval of cosine similarity. $\mathcal{M}_{(n-1)}$ is stored experience from previous $(n-1)$ episode.

4.3.2. Visual Scene Experience

Scene Experience Construction. During navigation, we extract visual descriptions at each step using the LLM, which are projected into the embedding vector d and stored in the scene-description experience bank $\mathcal{D}$. Meanwhile, visual appearance features are extracted using CLIP, and their embeddings f are stored in the scene-appearance experience bank $\mathcal{F}$. For the i -th navigation step in the n -th episode, the experience banks are constructed as follows:

$$\mathcal{D}_n, \mathcal{F}_n = \mathcal{D}_{n-1} \cup [(d_n^i)]_{i=1}^{i-1}, \mathcal{F}_{n-1} \cup [(f_n^i)]_{i=1}^{i-1}. \quad (10)$$

Please Note that experience can only be collected after a navigation step is completed. In other words, when the agent is executing n -th episode at i -th step, $\mathcal{D}$ or $\mathcal{F}$ stores only the scene images before n -th episode at $(i - 1)$ step.

Experience Retrieval. Here, we use the future landmarks $\mathcal{L}^f$ as the query text, which response potential direction selection (cf. Eq. 6), to retrieve scene experience in two ways: text-description retrieval ($T2D_{retrieval}$) and text-visual retrieval ($T2V_{retrieval}$). Specifically, the former focuses on the text-to-description similarity in query text and scene description d , while the latter focuses on the text-to-visual similarity by using CLIP [37] to project query text and stored visual images into feature space. For the n -th episode at i -th step, the retrieved top-K scene images $\mathcal{V}_n^e$ (cf. Figure 5 (c) about K-number ablation) is formulated as,

$$\mathcal{V}_n^e = \mathcal{V}^d + \mathcal{V}^f, \quad (11)$$

$$\mathcal{V}^d = T2D_{retrieval} \left(\frac{\mathcal{L}_i^f \cdot \mathcal{D}_n}{\|\mathcal{L}_i^f\| \cdot \|\mathcal{D}_n\|} \right), \quad (12)$$

$$\mathcal{V}^f = T2V_{retrieval} \left(\frac{\mathcal{L}_i^f \cdot \mathcal{F}_n}{\|\mathcal{L}_i^f\| \cdot \|\mathcal{F}_n\|} \right). \quad (13)$$

4.4. Action Prediction

Based on the above process, the agent’s action prediction is improved by historical experience and future thought, as

- **View Observation:** Concisely describe visible objects in the view and their spatial relationships.
- **Thought Assistance:** Using $\mathcal{L}^f$ in assisting current direction prediction. *Please see thought details in Eq. 6.*
- **Experience Assistance:** Retrieve the related trajectory $\mathcal{E}_n^t$ and scene $\mathcal{E}^v$ experience for the current episode. *Please see experience details in Eq. 9 and Eq. 11.*
- **Viewpoint selection:** Based on the above information, select the most suitable viewpoint to move.

Overall, for i -th navigation step at the n -th episode, the agent’s action prediction is formulated as,

$$\mathcal{R}_n, \mathcal{S}_n, \mathcal{V}_n = \mathcal{LLM}(\mathcal{I}^s, \mathcal{I}^t, \mathcal{I}, \mathcal{H}, \mathcal{O}, \mathcal{T}^e, \mathcal{V}^e, \mathcal{A}^d), \quad (14)$$

where $\mathcal{R}_n$, $\mathcal{S}_n$, and $\mathcal{V}_n$ denotes the agent’s reasoning process, viewpoint description, and selected viewpoint to move. Finally, we aggregate that information with the previous $\mathcal{H}$ to update the n -step navigation history $\mathcal{H}'$.

5. Experiments

5.1. Experiment Setup

Simulated Environments. Following the protocol of previous zero-shot works [35, 38, 39, 54], we use the sampled R2R-CE [6] for evaluation. Following the above protocol, we sample a new subset from more complex NavRAG-CE [48] for a better evaluation, which requires the agent to understand difficult human demands. *Please see supplementary material for more details about the datasets.*

Real-world Environments. To further evaluate our approach in real-world settings, we conducted comprehensive experiments across four representative indoor environments: Laboratory, Corridor, Elevator Lobby, and Study Room. For each, we collected and annotated 20 language instructions, covering both simple and complex tasks. Note that these annotated samples are excluded from the simulated datasets. This diverse set allows for a thorough evaluation of the agent’s performance in real-world conditions.

Evaluation Metrics. Following the standard VLN-CE protocol [4, 5, 38, 39, 54], we evaluate our EvoNav navigation performance in the following metrics: success rate (SR), oracle success rate (OSR), normalized Dynamic Time Warping (nDTW), success weighted by path length (SPL), trajectory length (TL), and navigation error (NE).

5.2. Comparison on Simulated Environment

Table 1 and Table 2 compare the performance of EvoNav with recent VLN-CE methods on the R2R-CE and NavRAG-CE benchmark, including supervised VLN-CE, zero-shot VLN-CE with extra pre-training, and zero-shot VLN-CE with train-free approaches. Overall, compared to zero-shot VLN-CE methods on Table 1 and Table 2, our approach achieves state-of-the-art (SOTA) results across all metrics, demonstrating the effectiveness of EvoNav. All methods, except Smartway-GPT4o [38] and MapGPT-CE-GPT4o [11], are evaluated using the same waypoint predictor to allow us to exclude and focus on the performance of the methods themselves. Specifically, in Table 1, EvoNav outperforms Open-Nav with the same Gemini-2.5-pro on SR (+20%) and OSR (+21%), and outperforms Smartway-GPT4o [38] of pretraining the waypoint predictor with more data on NE, SR, and SPL. In Table 2, EvoNav also outperforms other recent methods and achieves SOTA results across multiple diverse LLMs on NavRAG-CE, which tests the agent’s ability to handle more complex user demands. This proves the superiority and effectiveness of our EvoNav.

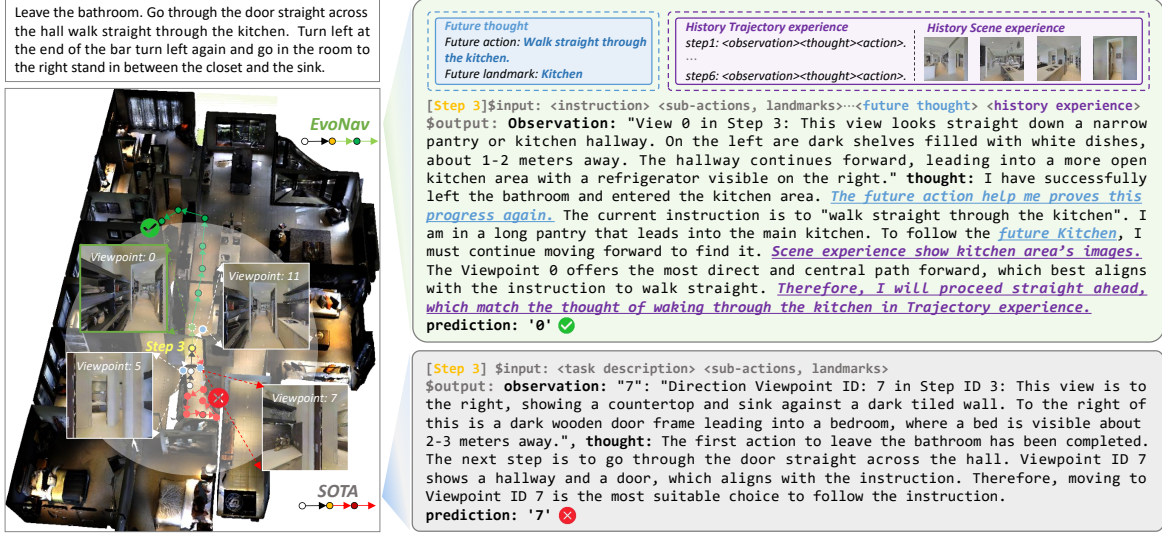


Figure 3. Navigation demo in simulated environment on R2R-CE. The left denotes navigation trajectories between the SOTA method and our EvoNav. For a better view, we also provide the navigation decision-making details of the key step, which highlight the effect of the proposed future thoughts and historical experience. Please see more navigation demo of the simulated environment in the Appendix.

Table 1. Performance comparison on the R2R-CE benchmark. “†” denotes using the result reported in [38]. “-” denotes unavailable statistics. **Best** and second-best results are highlighted.

Method	NE↓	nDTW↑	OSR↑	SR↑	SPL↑
<i>VLN-CE with supervised learning</i>					
CMA [20]	6.92	50.77	45	37	32.17
RecBERT[20]	5.8	54.81	57	48	43.22
BEVBert [4]	5.13	61.40	64	60	53.41
ETPNav [5]	5.16	61.16	63	58	52.19
<i>Zero-shot VLN-CE with extra pre-training</i>					
SmartWay-GPT4o [38]	7.01	-	51	29	22.46
<i>Zero-shot VLN-CE with train-free</i>					
Random	8.63	34.08	12	2	1.50
LXMERT [20]	10.48	18.73	22	2	1.87
DiscussNav-GPT4† [29]	7.77	42.87	15	11	10.51
NavGPT-CE-GPT4† [57]	8.37	-	26.9	16.30	10.20
MapGPT-CE-GPT4o† [11]	8.16	-	21	7	5.04
Open-Nav-GPT4o [35]	6.70	45.79	23	19	16.10
Open-Nav-Gemini-2.5-pro [35]	7.28	49.51	30	23	19.90
EvoNav-GPT4o (ours)	<u>5.97</u>	<u>56.32</u>	<u>35</u>	<u>30</u>	<u>24.91</u>
EvoNav-Gemini-2.5-pro (ours)	5.04	62.38	51	43	37.77

5.3. Comparison on Real-world Environments

To further evaluate the generalizability of our approach, we deploy EvoNav on an omnidirectional wheeled robot in real-world environments. We collect data from four diverse scenes, *e.g.*, Laboratory, Corridor, Elevator Lobby, and Study Room, annotating 20 samples per scene. As shown in Table 3, we compare our method with several previous works (*e.g.*, CMA, BevBert, and Open-Nav) in the zero-shot VLN-CE setting, achieving SOTA results across all metrics. Additionally, we visualize several real-world

Table 2. Comparison with recent SOTA methods on the NavRAG-CE benchmark. **Best** and second-best results are highlighted.

Method	NE↓	OSR↑	SR↑	SPL↑
<i>Zero-shot VLN-CE with train-free</i>				
Random	14.95	9.15	2.13	1.76
InstructNav-GPT4o [30]	13.62	20.62	13.40	7.93
DiscussNav-GPT4o [29]	7.99	22.73	14.58	8.25
Open-Nav-GPT4o [35]	6.81	38	22	11.89
Open-Nav-Gemini-2.5-pro [35]	<u>6.77</u>	<u>44</u>	24	13.39
EvoNav-GPT4o (ours)	<u>6.77</u>	41	29	14.28
EvoNav-Gemini-2.5-pro (ours)	6.20	51	30	17.60

navigation demos in Figure 4. Overall, these results prove the effectiveness of our method for real-world navigation.

5.4. Ablation Study

5.4.1. Ablation of Different LLMs on Navigation

As shown in Table 5, we generalize EvoNav across both open-source and closed-source LLMs. Overall, agents using closed-source LLMs (*e.g.*, Gemini-2.5-pro) outperform those based on open-source models, benefiting from richer data optimization and computing resources. Despite this, our EvoNav that equips open-source models, *e.g.*, Llama4-Maverick, achieves comparable SR and SPL to GPT-4o. This demonstrates that EvoNav can effectively adapt to different LLMs and unleash their reasoning capabilities, proving its robustness and generalizability.

5.4.2. Ablation of Key Components of EvoNav

(1) **A1→A3: Future Action and Landmark in F-CoT.** We analyze the impact of future action (*cf.* Eq. 4) and landmark (*cf.* Eq. 6) information in F-CoT, as shown in A1→A3 of Table 4. Here, “Base” denotes results of the

Table 3. Comparison on Real-world Environments

Method	Laboratory			Corridor			Elevator Lobby			Study Room		
	NE↓	OSR↑	SR↑	NE↓	OSR↑	SR↑	NE↓	OSR↑	SR↑	NE↓	OSR↑	SR↑
CMA[20]	2.94	30	30	3.11	20	20	3.01	20	10	3.10	25	20
BEVBert[4]	3.00	25	20	3.05	30	25	2.96	25	20	3.12	35	20
NaVid[54]	2.88	35	30	3.06	25	20	2.74	30	25	3.03	25	20
Open-Nav-Gemini-2.5-pro [35]	2.86	35	35	3.07	20	15	2.72	30	25	3.05	25	15
EvoNav-Gemini-2.5-pro (ours)	2.58	45	40	2.75	40	35	2.55	40	35	2.66	45	40

Table 4. Ablation of key components of F-CoT and H-CoE of EvoNav on R2R-CE Dataset.

Methods	F-CoT (Sec. 4.2)		H-CoE (Sec. 4.3)			NE↓	nDTW↑	OSR↑	SR↑	SPL↑
	$\mathcal{P}^f$ (Eq. 4)	$\mathcal{L}^f$ (Eq. 6)	$\mathcal{T}^e$ (Eq. 9)	$\mathcal{V}^d$ (Eq. 12)	$\mathcal{V}^f$ (Eq. 13)					
Base	$\times$	$\times$	$\times$	$\times$	$\times$	8.43	38.42	14	9	7.46
<i>Ablation of using future action $\mathcal{P}^f$ and landmark $\mathcal{L}^f$ in F-CoT</i>										
A1	✓	$\times$	$\times$	$\times$	$\times$	6.31	53.62	33	25	22.10
A2	$\times$	✓	$\times$	$\times$	$\times$	6.39	52.57	31	23	19.96
A3	✓	✓	$\times$	$\times$	$\times$	5.79	56.22	36	27	24.92
<i>Ablation of using trajectory $\mathcal{T}^e$ experience in H-CoE</i>										
B1	✓	✓	$\times$	$\times$	$\times$	5.79	56.22	36	27	24.92
B2	✓	✓	✓	$\times$	$\times$	5.69	58.10	42	31	27.64
<i>Ablation of using scene ($\mathcal{V}^d, \mathcal{V}^f$) experience in H-CoE</i>										
C1	✓	✓	$\times$	✓	$\times$	5.39	59.53	48	39	33.21
C2	✓	✓	$\times$	$\times$	✓	5.42	59.82	46	37	32.78
C3	✓	✓	$\times$	✓	✓	5.36	60.09	48	40	34.07
<i>Ablation of using trajectory $\mathcal{T}^e$ and scene ($\mathcal{V}^d, \mathcal{V}^f$) experience in H-CoE</i>										
D1	✓	✓	✓	$\times$	✓	5.37	60.24	47	39	33.39
D2	✓	✓	✓	✓	$\times$	5.05	61.11	51	41	36.11
Full pipeline	✓	✓	✓	✓	✓	5.04	62.38	51	43	37.77

Table 5. EvoNav performance with different LLMs on R2R-CE.

Method	NE↓	nDTW↑	OSR↑	SR↑	SPL↑
<i>The close-source LLMs</i>					
GPT-4o	5.97	56.32	35	30	24.91
Gemini-2.5-pro	5.04	62.38	51	43	37.77
Claude-sonnet-4.5	7.77	45.67	23	18	14.98
Doubao-1.5-Thinking-vision-pro	6.44	53.67	36	28	23.26
<i>The open-source LLMs</i>					
Qwen2.5-vl-72b	6.78	49.29	24	19	15.76
GLM-4.5v	7.13	45.96	21	16	13.89
Qwen3-vl-235b-a22b-instruct	6.86	51.25	37	30	23.89
Llama4-Maverick	6.79	49.28	31	27	22.43

baseline, which is direct answering through Gemini-2.5-pro from the visual observation and instruction. In view of “Base”, A1 outperforms A2. This is because the future landmarks obtained in A2 are sub-optimal, resulting from incomplete progress estimation due to the absence of future-action guidance. Based on A2, the performance of A3 is improved. This proves that both future action and landmarks are essential for better decision-making.

(2) B1→B2: Trajectory Experience in H-CoE. We evaluate the impact of trajectory experience (cf. Eq. 9), as shown in B1→B2 of Table 4. Compared to B1, B2 achieves a significant performance gain when provided with summarized experiences from similar past episodes. This highlights the effectiveness of historical trajectory experience.

(3) C1→C3: Scene Experience in H-CoE. We examine the effect of scene description (cf. Eq. 12) and appearance (cf. Eq. 13) experience in H-CoE, as shown in C1→C3 of

Table 4. Here, we removed the trajectory experience to allow us to focus on the effect of scene experience itself. Based on A3, agents with either scene description or appearance experience show performance gains, with the former achieving more. Compared with visual appearance in C1, C2 of visual description achieves higher performance, as LLMs with rich commonsense knowledge provide more refined visual observation even through using only a textual description. Nevertheless, visual appearance can still offer key visual information that is absent in LLM’s textual description, e.g., color, layout, and texture, improving the visual appearance. Finally, C3 achieves the best performance when combining both scene description and appearance.

(4) D1→D2: Trajectory and Scene Experience in H-CoE. We examine the effect of combining trajectory, scene description, and appearance experience, as shown in D1→D2 of Table 4. Similar to the findings on the strength of visual description in LLMs, D2 outperforms D1, which combines historical textual trajectory and holistic appearance experience. The full pipeline achieves the best performance by integrating F-CoT and H-CoE. In summary, the ablation study confirms that all components of EvoNav are essential and contribute synergistically to the final performance.

5.4.3. Ablation of Retrieval Period

We evaluate the performance of different experience retrieval periods for VLN-CE, as shown in Figure 5(a). “0-

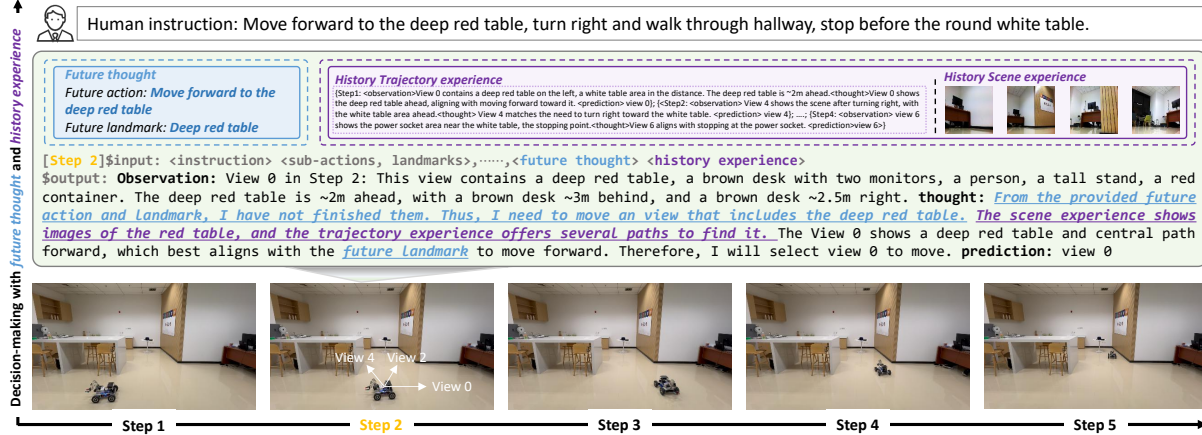


Figure 4. Navigation demo in the real world. For a better view, we also provide the decision-making details of the key step (step 2), which highlights the effect of the proposed future thoughts and historical experience. *Please see more real-world demos in the Appendix.*

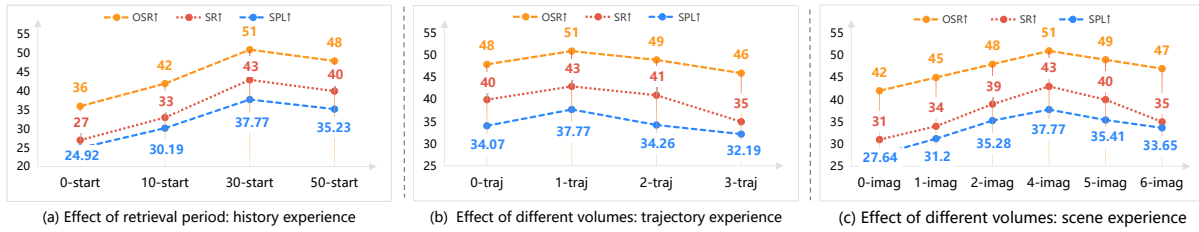


Figure 5. Ablation of different retrieval period (a), volumes of trajectory experience (b), and volumes of scene experience (c).

start” refers to not using historical experience (cf. A3 of Table 4), while “10/30/50-start” indicates the agent begins constructing the experience bank during the first 10/30/50% of the navigation episode and starts retrieving experience afterward. The results show performance improves gradually, peaking at 30%, then declining. This suggests that starting retrieval too early leads to lower similarity between the current episode and retrieved experiences, as the experience bank contains fewer relevant examples. Conversely, starting retrieval too late limits the benefit of the experience.

5.4.4. Ablation of Experience Volume

As shown in Figure 5 (b)→(c), we evaluate the performance of different volumes of trajectory and scene experience. “0-traj/image” means no trajectory/scene experience is used from the full pipeline (cf. C3/B2 of Table 4). “1/2/3-traj” and “1/2/3/4/5/6-image” refer to retrieving corresponding numbers of trajectory/scene experiences. In Figure 5 (b), results of “1-traj” outperforms “2-traj”, achieving the best results. Similarly, in Figure 5 (c), scene experience performance improves gradually, peaking at “4-image”. This suggests that increasing historical experience does not necessarily improve performance. Excessive experience, however, can introduce irrelevant noise, reducing context quality and, ultimately, navigation performance.

5.4.5. Analysis of Run Time and GPU Memory

Table 6 summarizes the computational cost comparisons between EvoNav with the more recent Open-Nav. For a fair comparison, we fixed them with the same GPU device and

Table 6. Comparisons of run time and GPU memory on the same RTX 3090 GPU and the same LLM’s API (Gemini-2.5-pro).

Method	OSR↑	SR↑	Time (mins) ↓	GPU (GB) ↓
Open-Nav [35]	30	23	14.7	12.5
EvoNav (ours)	51 _{↑21}	43 _{↑20}	7.6 _{↓7.1}	2.4 _{↓10.1}

LLM’s API (Gemini-2.5-pro). Specifically, our EvoNav not only outperforms to SOTA method over **20%** on SR and OSR but also is more efficient, such as Open-Nav vs. Ours: 1) inference GPU memory: 12.5 G vs. 2.4 G; 2) inference times per episode: 14.7 mins vs. 7.6 mins. Overall, this proves our work is more effective and also practical.

6. Conclusion

This paper rethinks LLM-based zero-shot VLN-CE and introduces a framework with the history-future reasoning (named EvoNav) for zero-shot VLN-CE, which improves the agent’s decision-making by history experience and future thought via Future Chain-of-Thought (F-CoT) and History Chain-of-Experience (H-CoE). F-CoT predicts future actions and landmarks as thoughts to assist navigation progress estimation and direction selection, while H-CoE summarizes historical trajectories and scenes as experience to improve navigation decision reliability. In future work, we aim to explore the multi-agent collaboration to improve the reasoning ability of LLM-based agents. One more thing, we hope EVONAV can encourage the embodied community to further explore the idea of “*review history and dream future*” in evolving challenging agents’ decision-making.

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